Nine Percent Labelled: What Has to Be True Before Self-Supervision Helps

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Abstract

We set out to apply self-supervised pretraining to nurse stress detection, on a dataset where 90.9% of recorded sensor time carries no label at all. The appeal was obvious: two hundred times more unlabelled than labelled data, and a natural pretext task. We never got there, and this paper is an account of why that is the more useful result. Pretraining requires two things this dataset does not supply. It requires a negative class, and no participant ever recorded being unstressed: all 358 survey entries mark an episode a nurse flagged, and the rating is its severity. It also requires a baseline worth improving on. Establishing both consumed the study. We show that the labels do mark a physiological state, but in one channel only: skin conductance rises at reported onset in 64.3% of episodes ($p = 0.0009$), while heart rate is indistinguishable from chance ($p = 0.80$). We construct a negative class by activity-matched sampling and verify by falsification that the resulting classifier detects neither movement nor shift schedule (both AUC 0.51). The baseline reaches AUC 0.765, far above a within-subject permutation null (0.498 ± 0.010), yet recovers only 10.6% of episodes at one false alarm per worn hour. A linear model reaches within 0.015 AUC of a boosted one, so capacity is not the constraint; three treatments of the labels, including non-negative positive-unlabelled risk estimation, all fail to improve on it. We stopped at a threshold fixed in advance. Two findings run against expectation: careful negative construction improves rather than inflates performance, and a published accuracy figure on this dataset is difficult to reconcile with subject-grouped evaluation.

1 Introduction

Occupational stress among nurses is associated with burnout, turnover, and adverse patient events. Wrist-worn sensors offer a continuous alternative to retrospective self-report, and a public dataset exists for studying it: Hosseini et al. (2) monitored 15 nurses for one week each during the COVID-19 outbreak, recording electrodermal activity, heart rate, skin temperature, and acceleration, with periodic smartphone-administered stress reports.

The dataset has a property that makes self-supervision attractive. Only 9.1% of the 1251.7 hours of recording falls inside a rated episode. The remaining 90.9% is unlabelled physiology, which is exactly the regime where masked-reconstruction pretraining is supposed to help: learn what ordinary physiology looks like from everything, then fine-tune on the little that is labelled. That was our plan.

Two prerequisites stopped it. Self-supervision does not manufacture a negative class, and this dataset has none: every one of the 358 survey rows marks an episode a participant flagged, and the number attached is its severity, not its presence. Nor does self-supervision have anything to improve on until a baseline exists. Building both turned out to be the study.

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We therefore report the groundwork rather than the method we intended, and we organise it as a sequence of eliminations against one question:

What would have to be true for representation learning to help here?

Section 3 asks whether the labels mark anything physiological. Section 4 asks whether the comparison group we build leaks a confound. Section 5 asks how far standard features and a standard learner get, and whether the pipeline manufactures its own signal. Section 6 asks whether a different treatment of the labels helps. Each answer narrows what remains.

Contributions.

1. An event-triggered analysis showing that only one of four channels responds at reported onset, which reorders the feature priorities for this dataset.
2. A negative-class construction with falsification probes, and the finding that careful construction *improves* performance rather than inflating it, contrary to the usual warning.
3. A decision rule fixed before results were seen, governing whether to escalate to representation learning, together with the measurement that answered it and an explicit statement of what that measurement does not settle.
4. Two controls, a second learner and a within-subject permutation null, that separate what the data supports from what the pipeline produces.

2 Data and units of analysis

2.1 Source

We use the original Dryad deposit rather than the merged convenience files in common circulation. The merged version resamples every channel onto a shared grid, which discards the blood volume pulse and inter-beat interval streams and removes the only route to heart rate variability. Channels are read at their native rates: acceleration at 32 Hz, blood volume pulse at 64 Hz, electrodermal activity and temperature at 4 Hz, and heart rate at 1 Hz, taken from the file headers. Published descriptions giving 72 Hz and 10 Hz for pulse and temperature disagree with the files themselves, and we follow the files.

The archive contains 609 sessions totalling 1251.7 hours across 250 person-days. The median session is 1.17 hours rather than a full shift, so a participant's week is fragmented across roughly 40 short recordings.

2.2 Clock alignment

Sensor timestamps are UTC; survey timestamps are bare local wall-clock with no zone recorded. Choosing wrongly places nearly every episode outside its own recording and raises no error, so we resolved it empirically by counting how many rated episodes start inside a session belonging to the same participant. Treating them as UTC places 50 of 245 (20.4%); as daylight-saving-aware local time, 212 (86.5%). Thirty-three rated episodes (13.5%) fall outside every session even so, and are set aside.

2.3 Window, label, and the metric

The unit of prediction is a 120 s window with a 60 s hop. Two arguments fix this length. No labelled episode is shorter than 120 s, and a 300 s window would discard 15 episodes outright. Independently, ultra-short heart rate variability work finds 120 s sufficient for time-domain measures (5), with root mean square of successive differences validated from recordings as short as 15 s (6); the classical five-minute figure is a recording convention rather than a requirement.

A window is **positive** if at least half of it falls inside a high-severity episode. A window is **negative** only if it survives the construction in Section 4. Everything else is **excluded**, not set to zero.

Table 1: Event-triggered response at reported onset, 140 episodes with usable coverage. Robust statistics only; see the note below on why.

Channel	Median Δz	Episodes positive	Sign test	Median raw shift
Electrodermal activity	$+0.568$	90/140 (64.3%)	$p = 0.0009$	$+0.64$
Heart rate	-0.067	68/140 (48.6%)	$p = 0.80$	$+0.47$ bpm

The primary metric, defined before use. We report *episode-level recall at a fixed alarm rate*, and both halves need stating.

An **episode is detected** if at least half of its constituent windows fire. The fraction is fixed at one half in advance; we do not tune it.

The **alarm rate** is expressed per hour of worn time, not per hour of sampled negative time. These differ substantially. Our scored sample is 43.6% positive by construction, whereas true window prevalence over all worn windows is 6.49%, so counting alarms against the sampled negatives would understate the deployed rate roughly sevenfold. We therefore fix the operating point on the false positive rate, which is prevalence-invariant, and translate: at 6.49% prevalence there are 28.1 negative windows per worn hour, so one alarm per hour corresponds to a false positive rate of 0.0356. All operating points below are reported this way.

2.4 The structural problem

The survey contains 358 rows, of which 245 carry a severity rating, split 46 low, 20 medium, 179 high. **None of them means "not stressed."** Every row is an episode a participant actively flagged, and the rating is its severity. The majority labelled class is therefore the most severe one, and nine of fifteen participants have no medium-severity episodes at all.

Two consequences follow. Three-class modelling is unworkable, so we collapse to high-severity against a constructed baseline. And the comparison group does not exist in the data, making this a positive-unlabelled problem (1) rather than ordinary binary classification.

2.5 Exclusions and cohort

Exclusions are applied in a declared order, because they do not commute: measuring session length before removing non-wear time retains 517 sessions, and measuring it after retains 504. Thirteen sessions turn on rule order alone.

We remove short sessions, non-wear windows, sessions with a dead electrodermal sensor, unrated episodes from both classes, duplicate survey rows, and low- and medium-severity episodes; twelve overlapping episode pairs are merged to their outer span.

We further remove five participants. One has no eligible comparison windows at all. Four have median skin conductance between 0.07 and 0.11 μS, at or below the sensor floor. A low median alone does not disqualify a participant, since a small but varying signal remains informative after normalisation. The deciding statistic is relative variation, the interquartile range divided by the median: these four score 0.44 to 0.72, while the remaining eleven score 1.04 to 5.14, with no overlap. Their signal is flat, not merely small.

The retained cohort is 10 participants, 24,334 windows, and 138 high-severity episodes with sensor coverage. Three counts describe this dataset and only two are denominators: the window count is a compute statistic, the honest denominator for detecting an episode is 138, and for generalising to a new person it is 10.

3 Do the labels mark anything physiological?

Before building a classifier we ask whether there is anything to classify. We align every high-severity episode at its reported onset, normalise each channel against the participant's own preceding hour, and compare the first ten minutes after onset against the preceding thirty.

Table 1 gives the result. Skin conductance rises at onset in 64.3% of episodes, a shift that survives both a sign test and a Wilcoxon signed-rank test ($p = 3.3 \times 10^{-6}$), and the raw median shift of 0.64 μS sits on a baseline of 0.68 μS, roughly a doubling. Heart rate rises in 48.6% of episodes, which is indistinguishable from chance.

Why we report medians. The pooled mean tells a different and false story. It reports an electrodermal shift of $+9.75$ normalised units, of which 63% is contributed by five episodes out of 140. The cause is numerical rather than physiological: the trailing interquartile range used as a denominator approaches zero during flat stretches, and the largest single value reached $+1600$ where a plausible score is near ± 3. We bound the denominator at its empirical first percentile, clip the output, and fix the use of medians and sign tests in advance, because the same data yields opposite headlines otherwise.

This analysis is descriptive and uses all episodes; it informs which channels enter the feature set but is not itself a predictive claim, and no model is selected on it.

The result reorders the rest of the study. The problem is well posed, and it is well posed in one channel. Any subsequent result driven by heart rate deserves suspicion until explained.

4 Constructing a comparison group

4.1 Eligibility

A candidate negative window must satisfy four conditions: it lies in a session of at least thirty minutes; it falls on a participant-day containing at least one reported episode; it lies at least thirty minutes from the boundary of any reported episode, including unrated ones; and it is not itself positive.

The second condition carries most of the argument. On a day when a participant filed nothing, silence is uninformative: she may have been calm, or too busy to open the application. On a day when she filed at least one report, she was demonstrably engaging with the survey, so silence during the remainder of that day is evidence rather than absence of evidence. This condition is also the most expensive, reducing 24,334 windows to 16,214. All four together leave 7,955 candidate windows, or 265.2 hours.

4.2 Matching rather than filtering

Nurses move more when busy, and busy periods are stressful, so movement correlates with stress without being a stress signal. Selecting quiet periods as negatives would produce two classes differing in both stress and movement, and a classifier would separate them by detecting movement.

We therefore match rather than filter. We bin the positive class into deciles of mean acceleration magnitude, and for each bin draw negatives from the same bin, *within the same participant*. Global matching would draw negatives disproportionately from high-coverage participants, making participant identity a partial label proxy along exactly the axis the evaluation holds out.

4.3 The negative ratio

We fix the negative-to-positive ratio uniformly rather than letting each participant supply its maximum, because class balance determines where the decision threshold belongs. Table 2 makes the point concretely: one fixed model, recall 0.80 and specificity 0.80 throughout, scored on two folds differing only in prevalence.

Exact uniformity is unattainable. The binding constraint is a single cell, one participant's highest-activity bin containing 13 positives and 7 candidates, so matching every participant exactly would require a ratio below one. We report a sensitivity analysis across four arms in Table 3 and fix ratio $= 1.31$, which halves the residual spread from $1.64\times$ to $1.28\times$. We note that the alternative arm at 2.14 yields marginally higher AUC; we prefer 1.31 because comparable per-fold thresholds matter more here than a difference of 0.002, and we report both rather than choosing silently.

Negatives are drawn under three random seeds. The seed-to-seed spread is the noise floor: any later comparison smaller than it is not a comparison.

Table 2: The same model on 100 windows at two prevalences. Recall and specificity are held fixed; only the class balance changes. F_1 falls from 0.727 to 0.195 because the false positives are drawn from a much larger negative pool.

	2:1 fold (33.3% positive)	32:1 fold (3.0% positive)
Precision	0.667	0.111
Recall	0.800	0.800
F_1	**0.727**	**0.195**

Table 3: Falsification probes and ratio sensitivity, mean of three seeds. All probes pass in all arms. Area under the ROC curve is prevalence-invariant, so arms with different denominators remain comparable.

Arm	Participants	Balance spread	ACC only	Time only	EDA only	Full
All 10, ratio 1.31	10	1.28×	0.513	0.507	0.768	0.749
All 10, ratio 2.14	10	1.64×	0.511	0.510	0.770	0.736
Drop one, ratio 1.31	9	1.07×	0.521	0.533	0.762	0.739
Drop one, ratio 2.25	9	1.30×	0.517	0.525	0.765	0.726

4.4 Falsification

We then build models we want to fail. Each is given only one suspicious signal, using the same learner as the baseline so that a null result is attributable to absence of signal rather than to a weaker model.

A model given only acceleration reaches AUC 0.513, and one given only the hour of day reaches 0.507, against a chance value of 0.500. Both hold in all four arms. The constructed comparison group therefore does not leak movement or shift schedule.

Participant identity is recoverable from the same features at 0.768 accuracy against a chance value of 0.100. We record this as a measurement rather than a failure, since physiology genuinely differs between people; it is the reference against which any learned representation would have to be compared.

5 Baseline and controls

Features are extracted at native rates and then aggregated to the window. Detecting skin conductance responses on a 1 Hz downsample loses roughly 40% of them, so the common 1 Hz grid is used only for the label join and coarse features. The feature set comprises tonic level and slope, phasic variability, response count and mean amplitude for electrodermal activity; mean, variability, slope, maximum and deviation from a trailing median for heart rate; magnitude statistics and a movement fraction for acceleration; slope and deviation from a trailing median for temperature, never its absolute value, which has an intraclass correlation of 0.52 with participant identity and acts as a fingerprint; and hour of day with position in session.

Heart rate variability is excluded from the baseline. Although the inter-beat interval file is present in every session, only 5.9% of 120 s windows contain a run of clean consecutive beats long enough to compute it, and the archive holds only 39.5 hours of usable beat data out of 1251.7 hours. Reported session-level availability of 48% is eight times more flattering than the window-level figure that governs a per-window feature.

Continuous channels are normalised against a trailing sixty-minute robust baseline within session, which uses only past data and is therefore both leakage-free and reproducible at deployment. The model is gradient boosting with balanced class weights, evaluated leave-one-participant-out across three negative-sampling seeds.

The model attains AUC 0.7654 with a seed range of 0.0095. Table 4 translates this into operating points.

Table 4: Baseline operating points. The alarm rate follows the translation in Section 2.3. The seed range on episode recall at one alarm per hour is comparable to the estimate itself, and we quote it wherever the estimate appears.

Alarms per worn hour	False positive rate	Window recall	Episode recall	Seed range
0.5	0.018	0.087	0.029	0.000
1.0	0.036	0.155	**0.106**	0.072
2.0	0.071	0.292	0.280	0.058

Table 5: Controls. The permutation null shuffles labels within participant, preserving class balance, fold structure and feature distributions while destroying only the label-feature correspondence.

	AUC	Episode recall at 1 alarm/hour
Gradient boosting	0.7655	0.1063
Logistic regression	0.7502	0.0942
Permutation null (mean of 20)	0.4975 (±0.0099)	—
Permutation null (maximum)	0.5186	—

The gap is the finding. An AUC of 0.765 means the model reliably ranks stressed windows above unstressed ones. At one false alarm per worn hour it recovers 10.6% of episodes, with a seed range of 0.072 that is comparable to the estimate itself. AUC integrates over false positive rates including regions no deployment would tolerate; once constrained to the usable corner, little remains. Two of ten participants detect zero episodes at any usable threshold, and one is below chance at AUC 0.388. We report per fold and never average; fold AUC ranges from 0.388 to 0.885.

Compute. All experiments run on a single laptop CPU with no GPU. Feature extraction over 387 sessions takes 107 s; the complete set of reported experiments, including the twenty-draw permutation null and the four-arm ratio sensitivity, completes in under an hour. The audit that precedes them reads the 3.5 GB archive once per section and is likewise CPU-bound. No experiment reported here was preceded by a larger unreported sweep.

5.1 Two controls

Table 5 reports two controls that separate what the data supports from what the pipeline produces.

Shuffling labels within participant yields AUC 0.4975 ± 0.0099 across twenty permutations, with a maximum of 0.5186. The observed value exceeds every null draw, roughly twenty-six standard deviations above the null mean. The pipeline does not manufacture its own signal.

A penalised logistic regression on identical folds and features reaches AUC 0.7502, within 0.0153 of the boosted model. A linear model with almost no capacity to overfit arrives at nearly the same place, so model capacity is not the binding constraint.

6 Does a different treatment of the labels help?

We fixed a decision rule before computing the following, and record it here as written:

> Proceed to representation learning only if a different label treatment moves episode recall at one alarm per worn hour by more than the baseline seed range of 0.0724.

The threshold is the baseline's own measured seed range rather than a chosen constant. We note that this makes the rule generous in one direction: a noisier baseline sets a higher bar and is easier to leave uncleared.

We compare three treatments on identical folds, windows, and features: the curated negatives of Section 4; a naive comparator treating all unlabelled time as negative, with no same-day rule, guard band, or matching; and non-negative positive-unlabelled risk estimation (3) fitted participant-stratified,

Table 6: Label treatments on identical folds. Both reported metrics are prevalence-invariant, so the naive arm scored over 24,334 windows and the curated arm over 3,622 remain comparable. The positive-unlabelled rows use a fixed training budget without an inner validation split and are marked accordingly; they are not tuned to the same standard as the baseline.

Treatment	Class prior	AUC	Episode recall at 1 alarm/hour
Curated negatives (baseline)	—	**0.7654**	**0.1063**
Naive, all unlabelled negative	—	0.6925	0.0652
Non-negative PU (untuned)	0.05	0.6616	0.0435
Non-negative PU (untuned)	0.10	0.6933	0.0652
Non-negative PU (untuned)	0.20	0.7132	0.0507
Non-negative PU (untuned)	0.30	0.7167	0.0580

with the class prior bracketed rather than estimated, since it is not identifiable from positive-unlabelled data.

The threshold to clear was 0.1787. The best alternative reached 0.0652, a change of -0.0411. **The rule was not met, and every alternative is worse than the baseline rather than merely insufficiently better.**

Curation improved rather than inflated. This comparison is normally run to detect inflation: naive negatives include off-shift and sleeping time, the contrast becomes trivial, and the score rises. Here the curated construction beats naive by 0.073 AUC and 0.041 episode recall. The likely mechanism is the guard band, since naive negatives include windows adjacent to episodes, which are physiologically similar to positives and act as label noise during training. We have not isolated the guard band from the same-day rule, so we offer this as the plausible mechanism rather than a demonstrated one.

The prior does not rescue it. AUC rises monotonically with the class prior, from 0.662 to 0.717, which is the shape one could present as promising. Episode recall at the deployable operating point does not follow, remaining between 0.044 and 0.065 throughout. We report the prior-indexed results in full rather than the trend alone.

7 Discussion

7.1 What the eliminations leave, and what they do not

The labels mark a real physiological state, in one channel. The comparison group leaks neither movement nor schedule. The pipeline does not manufacture signal, since a within-participant permutation null sits at chance. Model capacity is not the constraint, since a linear model matches a boosted one. Three treatments of the labels do not improve on the baseline, and two make it worse.

What we have eliminated is the label-treatment axis, not the representation axis. Our decision rule tested whether a different way of *using* the labels helps. Self-supervised pretraining asks a different question: whether a different *representation* of the signal helps. The rule we fixed does not settle that, and we state this rather than treating the null as broader than it is.

What we can say is narrower and still useful. Within the feature set and learner space we searched, the results do not move; the constraint is not capacity, not the comparison group, and not the pipeline. The remaining candidates are the representation and the data. The data-side candidates are concrete: 138 episodes from 10 participants; labels that are retrospective self-reports of intervals, treated here as uniform states although a thirty-minute report is unlikely to describe thirty uniform minutes; a constructed rather than observed negative class; and one responsive channel out of four.

7.2 What this implies for the pretraining we did not run

A label-efficiency comparison remains the informative experiment, and it is worth stating what it would need. The measurement is not whether a pretrained encoder beats hand-crafted features outright, which is unlikely at 138 episodes, but whether it reaches a given level with fewer of them. That curve would separate a label-supply constraint from a representation constraint directly.

Two preconditions follow from our results. The participant-identity probe must be compared against 0.768, the accuracy reachable from sensible hand-crafted features, and not against chance; an encoder scoring near 0.95 would have learned identity rather than physiology. And pretraining on all participants before a leave-one-participant-out evaluation is transductive, so it must either be re-run inside each fold or be labelled as an upper bound.

7.3 On the reported accuracy in prior work

Mathur et al. (4) report weighted $F_1 = 0.99$ on this dataset. We do not reproduce this and think it is difficult to reconcile with participant-grouped evaluation. Adjacent windows in these recordings share raw samples and participant-specific baselines; a split that does not respect participant boundaries places near-duplicates on both sides. We are not in a position to diagnose another group's protocol from a published summary, and we note the discrepancy rather than attributing a cause. We report our own protocol in full so that the comparison can be made.

7.4 What a future collection would need

The actionable consequence concerns label supply. A collection targeting this task should record explicit non-stress intervals, since their absence makes every downstream number conditional on a construction; capture onset more finely than a retrospective interval; and prioritise electrodermal signal quality, since four of fifteen participants here produced a flat trace on the only responsive channel.

8 Limitations

The cohort is 10 of 15 female nurses at a single hospital during a pandemic, and external validity is correspondingly narrow. Negatives are constructed, so precision is not reported as a performance claim: when the model flags an unlabelled window we cannot distinguish a false alarm from an unreported episode. The class prior is not identifiable and is bracketed rather than estimated. Our non-negative positive-unlabelled implementation uses a fixed training budget without an inner validation split, so its negative result is weaker evidence than the naive comparator, which shares the baseline's learner and features and differs only in label treatment. We searched two learners and one feature set; a sequence model over raw windows is untested. Thirty-three rated episodes have no sensor coverage and are absent from every metric. Two normalisation constants, the denominator floor and the output clip, are choices we record and vary rather than defend. Ten folds is underpowered for the paired comparisons we report, and we state spreads descriptively rather than dividing them by the square root of the fold count. The permutation null uses twenty draws, so its p-value floor is 0.048; the observed value exceeds every draw but a finer resolution would require more permutations.

9 Broader impacts

The application is monitoring of employees during work, and that is worth naming plainly. A reliable stress detector deployed in a hospital could direct support toward staff who need it, which is the motivating case. The same instrument could as easily support surveillance, scheduling penalties, or insurance decisions, and the people measured are in a weak position to refuse. Consent to be monitored by an employer is not freely given in the way consent to a research study is.

Our result bears on this directly. A detector recovering one episode in ten at a deployable alarm rate is not fit for any of those uses, including the benevolent one, and two of ten participants are undetectable at any usable threshold. We would regard deployment on the strength of results at this level as harmful, and we note that a published figure of $F_1 = 0.99$ on this same dataset could be read as licensing exactly that. Reporting the achievable number honestly is the contribution most relevant to impact here.

The cohort is ten female nurses at one hospital during a pandemic. Any performance claim generalised beyond that population would be unsupported, and differential performance across groups cannot be assessed at this sample size.

10 Conclusion

We intended to test whether self-supervised pretraining helps when 90.9% of the data is unlabelled. Establishing the prerequisites consumed the study, and the prerequisites turned out to be the result. The labels mark a real state in one channel. A comparison group can be built that survives falsification. A baseline built on it clears a permutation null decisively and still recovers one episode in ten at a deployable alarm rate, and neither a different learner nor three different treatments of the labels move it.

We report this as a result rather than a failure, and we are explicit about its boundary: we have eliminated the label-treatment axis, not the representation axis. What a future study most needs from this one is not a score but a measurement of what was missing, and the most useful next experiment is a label-efficiency curve rather than another point estimate.

References

[1] C. Elkan and K. Noto. Learning classifiers from only positive and unlabeled data. In *Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2008. DOI: 10.1145/1401890.1401920.

[2] S. Hosseini, R. Gottumukkala, S. Katragadda, R. T. Bhupatiraju, Z. Ashkar, C. W. Borst, and K. Cochran. A multimodal sensor dataset for continuous stress detection of nurses in a hospital. *Scientific Data*, 9:255, 2022. DOI: 10.1038/s41597-022-01361-y.

[3] R. Kiryo, G. Niu, M. C. du Plessis, and M. Sugiyama. Positive-unlabeled learning with non-negative risk estimator. In *Advances in Neural Information Processing Systems 30*, 2017. arXiv:1703.00593.

[4] A. Mathur et al. Body sensor-based multimodal nurse stress detection using machine learning. In *2024 16th International Conference on Communication Systems and Networks (COMSNETS)*, 2024.

[5] M. L. Munoz, A. van Roon, H. Riese, C. Thio, E. Oostenbroek, I. Westrik, E. J. C. de Geus, R. Gansevoort, J. Lefrandt, I. M. Nolte, and H. Snieder. Validity of (ultra-)short recordings for heart rate variability measurements. *PLOS ONE*, 10(9):e0138921, 2015. DOI: 10.1371/journal.pone.0138921.

[6] M. Orini et al. Long-term association of ultra-short heart rate variability with cardiovascular events. *Scientific Reports*, 13, 2023.

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NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: blue[Yes]

   Justification: The abstract states the negative result and its boundary explicitly: we eliminated the label-treatment axis and not the representation axis, and Section 7.1 repeats this rather than generalising past it. The intended contribution, an account of the prerequisites for self-supervision on sparsely labelled physiological data, matches what is delivered. Every quantity quoted in the abstract appears with its seed range where one exists.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: blue[Yes]

Justification: Section 8 is a dedicated limitations section. It names the narrow cohort, the constructed rather than observed negative class and the consequent inestimability of precision, the unidentifiable class prior, the undertuned positive-unlabelled arm and why its negative result is weaker evidence than the naive comparator, the two-learner search space, the 33 episodes with no sensor coverage, the two normalisation constants chosen rather than derived, the underpowered ten-fold comparisons, and the resolution floor of a twenty-draw permutation test.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: gray[N/A]

   Justification: The paper contains no theoretical results. The one formal claim it relies on, that the class prior is not identifiable from positive-unlabelled data, is cited rather than proved.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: blue[Yes]

   Justification: The source dataset is public and cited. The paper specifies the window length and hop, the label rule and its threshold, the exclusion rules and the order in which they are applied, the four eligibility conditions for negatives, the matching procedure and its binning, the negative ratio and how it was chosen, the number of sampling seeds, the feature set by channel, the normalisation and its two constants, the learner, and the fold structure. Where a choice could have gone otherwise, the alternative is reported alongside it rather than omitted.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: orange[No]

   Justification: The dataset is public and cited by DOI. The analysis code is not released with this submission, because the repository would identify the authors and the review is double-blind. The protocol is specified in full in Sections 2 through 6 with the intent that it be reimplementable from the paper alone. Code will be released on acceptance or on publication elsewhere.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: blue[Yes]

   Justification: Sections 2, 4 and 5 give the splits (leave-one-participant-out, grouped on participant, with the reason stated), the learner and its class weighting, the normalisation window and its floor and clip, the negative ratio and the sensitivity analysis over it, and the class prior grid for the positive-unlabelled arm. Choices fixed before results were seen are identified as such, including the episode-detection fraction and the decision rule in Section 6.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: blue[Yes]

Justification: Every quantity computed on the constructed negatives is reported with its range across three negative-sampling seeds, and that range is what the variability captures: the stochastic draw of the comparison group, holding folds, features and learner fixed. The onset analysis reports a sign test and a Wilcoxon signed-rank test rather than a parametric interval, because the underlying distribution is heavy-tailed for reasons the paper explains. The baseline is compared against a twenty-draw within-participant permutation null, reported as mean, standard deviation and maximum. Where the seed range is comparable to the estimate itself, as it is for episode recall, the paper says so at every occurrence rather than only once. Fold-level results are reported individually and never averaged, since two of ten folds behave qualitatively differently from the rest.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: blue[Yes]

   Justification: A compute paragraph in Section 5 states that all experiments run on a single laptop CPU with no GPU, gives the feature-extraction time over 387 sessions, and states that the complete set of reported experiments finishes in under an hour. It also records that no reported experiment was preceded by a larger unreported sweep.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: blue[Yes]

   Justification: The work is a secondary analysis of a public, de-identified dataset whose original collection was ethically reviewed by the depositing institution. No new data were collected and no participant was contacted. Anonymity is preserved in this submission.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: blue[Yes]

    Justification: Section 9 addresses this. The positive case is directing support toward staff who need it. The negative case is that the same instrument supports surveillance, scheduling penalties, or insurance decisions, and that employees are poorly placed to refuse monitoring by an employer. The section argues that a detector performing at the level reported here is unfit for any of these uses including the benevolent one, and notes that a published figure of $F_1 = 0.99$ on this dataset could be read as licensing deployment that our results do not support.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: gray[N/A]

    Justification: No model, dataset, or scraped resource is released. The trained models are not distributed and, on the paper's own argument, are not fit for deployment.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: blue[Yes]

    Justification: The dataset is credited to its depositors and cited both as a Dryad deposit by DOI and as the accompanying data descriptor. The deposit landing page states that content is licensed for reuse but does not display a specific license designation, so we describe it as publicly available under the repository's terms rather than naming a license we cannot verify. Software used is open source: scikit-learn, NumPy, pandas, and NeuroKit2 for electrodermal response detection.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: gray[N/A]

    Justification: No new assets are released with this submission.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: gray[N/A]

    Justification: This is a secondary analysis of an existing public dataset. We did not recruit, instruct, compensate, or interact with any participant. Instructions given to the original participants and the survey instrument are documented by the depositors in the cited data descriptor.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: gray[N/A]

    Justification: No human subjects research was conducted for this paper. The original collection was reviewed and approved by the depositing institution's review board, as reported in the cited data descriptor; we rely on that record and did not seek separate approval for secondary analysis of de-identified public data.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: gray[N/A]

    Justification: No large language model is a component of the method. The pipeline consists of signal processing, feature extraction, and gradient boosting or logistic regression, none of which involve an LLM. A coding assistant was used while implementing the analysis and drafting the manuscript, which the question's own guidance places outside the declaration requirement; every reported number was produced by the scripts and verified against them, and every reference was checked against Crossref, arXiv, or a literature index rather than generated.